

CINCINNATI NORTHERN KENTUCKY, KY, USA

WMO: 724210

Lat: 39.044N

Lon: 84.672W

Elev: 269

StdP: 98.13

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-11.7	-20.0	0.7	-13.8	-17.1	0.9	-10.6	11.6	2.8	10.6	1.6	3.3	290	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	33.0	23.4	31.7	23.0	30.4	22.5	25.3	30.4	24.5	29.3	23.8	28.3	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.3	27.9	23.0	18.4	27.0	22.4	17.7	26.3	78.6	30.4	75.3	29.5	72.5	28.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.6	-19.1	34.6	4.3	2.0	-22.1	36.1	-24.6	37.3	-27.0	38.4	-30.2	39.9		
(5)			-19.5	26.6	4.1	0.8	-22.4	27.1	-24.9	27.6	-27.2	28.0	-30.2	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.6	1.5	6.4	12.7	17.9	22.4	24.3	23.9	20.2	13.6	7.0	2.1		
	DBStd	10.10	6.92	6.43	6.12	5.17	4.41	3.14	2.67	2.65	3.96	4.96	5.37	5.85		
	HDD10.0	1153	336	245	146	31	2	0	0	0	21	121	252			
	HDD18.3	2711	587	471	371	178	64	6	1	1	26	162	341	503		
	CDD10.0	2132	7	7	34	112	247	373	442	432	307	132	31	8		
	CDD18.3	648	0	0	2	9	51	129	184	175	83	15	1	0		
	CDH23.3	5438	0	1	13	72	387	1066	1623	1471	694	109	2	0		
	CDH26.7	1818	0	0	1	6	87	348	597	525	232	22	0	0		
(14) Wind	WSAvg	3.6	4.2	4.2	4.3	4.3	3.4	3.1	2.8	2.7	2.9	3.3	3.8	3.9		
(15) Precipitation	PrecAvg	1065	85	68	98	104	117	113	93	76	77	73	70	91		
	PrecMax	1576	170	142	262	289	259	249	215	138	165	160	170	207		
	PrecMin	730	37	22	33	33	11	41	23	12	12	23	18	26		
	PrecStd	179	31	36	47	60	66	46	46	38	47	41	37	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	20.3	25.1	27.5	30.6	33.3	35.0	35.0	33.7	29.1	22.8	18.5		
		MCWB	14.9	15.3	16.8	18.2	21.7	23.3	24.9	23.2	21.5	20.1	16.4	15.7		
	2%	DB	15.3	16.6	21.7	25.5	28.9	31.6	32.9	32.7	31.3	26.2	19.9	15.7		
		MCWB	13.2	12.5	14.6	17.0	20.5	22.9	24.1	23.2	21.7	19.0	14.9	13.2		
	5%	DB	12.8	14.1	18.8	23.5	27.5	30.2	31.4	31.2	29.4	24.0	17.8	13.4		
		MCWB	10.9	10.8	13.1	15.9	20.1	22.3	23.4	22.8	20.8	17.8	14.1	11.1		
	10%	DB	9.8	11.4	16.2	21.4	25.7	28.8	30.0	29.6	27.4	21.8	15.5	11.0		
		MCWB	7.4	7.8	12.0	15.0	19.3	21.7	22.9	22.3	20.1	16.9	12.1	8.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.3	17.7	20.1	23.2	25.6	26.7	25.8	24.4	22.4	17.9	16.3		
		MCDB	17.2	19.5	23.6	25.5	28.5	30.6	32.1	31.2	29.6	26.8	20.8	18.3		
	2%	WB	13.6	13.7	15.8	18.5	22.0	24.4	25.5	25.0	23.3	20.7	16.4	13.8		
		MCDB	15.2	15.7	19.9	23.0	27.2	29.3	30.7	30.1	28.2	23.8	19.1	15.2		
	5%	WB	11.2	11.1	14.4	17.1	21.1	23.5	24.7	24.2	22.4	19.1	14.4	11.4		
		MCDB	12.5	13.5	17.8	21.2	25.6	28.4	29.6	28.6	26.9	22.8	17.1	13.0		
	10%	WB	7.8	8.0	12.3	15.9	20.1	22.7	24.0	23.5	21.5	17.4	12.5	8.7		
		MCDB	9.4	11.0	15.9	20.0	24.2	27.3	28.4	27.5	25.5	20.9	15.2	10.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.0	10.0	11.2	10.5	10.3	10.0	10.3	10.9	10.6	9.5	7.7		
		MCDBR	9.8	12.2	13.1	13.1	11.9	12.0	11.4	11.8	13.0	12.5	11.6	10.2		
	5% WB	MCWBR	8.4	9.4	7.8	7.3	5.8	5.3	4.4	4.3	5.1	6.5	8.0	8.7		
		MCWBR	9.3	11.1	11.1	11.0	10.4	10.3	10.0	9.9	10.4	10.5	10.2	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.325	0.352	0.391	0.432	0.446	0.457	0.451	0.400	0.354	0.325	0.303		
		taud	2.482	2.395	2.387	2.281	2.203	2.207	2.183	2.185	2.307	2.444	2.482	2.495		
	Ebn at Noon		870	897	905	885	851	837	824	818	842	853	839	847		
		Edh at Noon	79	98	110	130	143	143	145	141	118	93	78	72		
(44) All-Sky Solar Radiation	RadAvg		1.81	2.58	3.63	4.87	5.48	6.13	5.89	5.48	4.50	3.25	2.21	1.54		
	RadStd		0.17	0.33	0.33	0.38	0.48	0.35	0.35	0.30	0.45	0.33	0.25	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+78
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+72

Nomenclature: See separate page